Nicolas **Comte**, Ph.D



Profile

R&D Engineer & Computer Science Ph.D. specializing in Deep Learning, computer vision in medical imaging.

Combining expertise in research, engineering, and Agile methodologies, I design and conduct deep learning, modeling, and simulation projects.

Contact details

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cometicon.github.io

Personal information

Citizenship French
Family Married, 2 children
Languages French (native), English (C1)

Skills

AI & Computer Sci. Deep learning, 3D mesh processing, 2D/3D Medical Imaging analysis, Mocap
Software Eng. C++, Python, Git, CI/CD, Agile methodology
Tools & Methodology PyTorch, Scikit-learn, R, Matlab, Sofa
Communication LaTeX, HTML, Gimp, Krita, Inkscape

Professional Experience

R&D Engineer at *Circle (France).* **2024–2026**
◇ Designed, trained, and deployed deep learning models for automated analysis of 3D intraoral scans
◇ Maintained continuous technology watch, designed architectures, and selected implementation methods.
◇ Built and deployed end-to-end MLOps pipelines for automated model retraining and deployment.
◇ Implemented CI/CD workflows for ML models and performed code reviews, preventing regressions and ensuring code quality.
◇ Optimized 3D data pipelines for commercial software integration.
◇ Acted as Agile facilitator and developer, handling task breakdown, value assignment, and prioritization.

R&D Engineer & PhD Candidate at *Inria, Anatoscope (France).* **2019–2023**
◇ Industrial R&D: Conducted joint research on medical image processing, 3D anatomical modeling, and simulation of scoliotic spines.
◇ Algorithm Design: Developed novel machine learning methods integrating motion capture data into 3D structural spine models.
◇ Impact & Awards: Published research (Google scholar) and won Best Poster Award in AI for Biomedical Imaging (IABM 2023).

Software Engineer at *Inria (France).* **2016–2018**
◇ Engineered and implemented fast, open-source C++/Python scientific applications for molecular phylogeny.
See Treerecs and Seaview 5 (> 300 citations).
◇ Co-supervision of 2 interns (engineering school and BTS vocational degree on Programming)
+ teaching at *INSA Lyon* C++ programming.

Education

Ph.D in Computer Science. *Université Grenoble-Alpes.* **2020-2023**
◇ Thesis title: *Learning scoliosis patterns using anatomical models and motion capture.*

Engineer in Bio-Informatics and Modeling. *INSA Lyon.* **2013–2016**

Bachelor in Bio-Informatics. *Université de Lyon.* **2010–2013**

Awards

Best Poster in artificial intelligence applied in biomedical imaging. *IABM 2023, Colloque National en Intelligence Artificielle en Imagerie Biomédicale.*

Gold medal and Best Composite Part. *IGEM 2014, International Genetically Engineered Machine competition.*

Hobbies

Graphics: infographics, illustration, digital drawing and painting.
Science communication: member of *Démesures 2017-2019* (France), animator, speaker in *Geek Touch 2018*, *Fête de la Science 2017, 2018*, ...
Photography: hyper/macrophotography @cometicon.macros
Programming: various side projects in simulation and artificial life....